DATABASE PROJECT PROPOSAL:

LIBRARY MANAGEMENT SYSTEM

Bachelor in Information Technology (Fall 24 Self Support)

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Project Proposal: Library Management Database System (LMS-DB)

Course: Database Management Systems

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1. Project Title

Library Management Database System (LMS-DB)

2. Project Overview

The Library Management Database System is designed to serve as the efficient, persistent data back-end for the Library Management Application (LMS). It aims to centralize all library operational data, ensuring integrity, consistency, and efficient retrieval through a structured Relational Database Management System (RDBMS). This system will manage books, library members, and all transactions (issuing, returning, fines).

3. Problem Statement (Data Focus)

The C++ application requires a robust method for data storage beyond simple text files, which suffer from poor scalability, lack of integrity constraints, and slow query performance. This project addresses the need for:

- Ensuring **data integrity** (e.g., preventing a book from being issued twice).
- Enforcing **relationships** (e.g., connecting a transaction to a valid Member and a valid Book).
- Facilitating **complex queries** (e.g., generating overdue reports, calculating total fines by member).
- Providing a **scalable foundation** for the future C++ application.

4. Database Design Objectives

1. To design a fully normalized database schema (3NF or higher) for optimal performance and redundancy reduction.
2. To implement the schema using an RDBMS, defining all Primary Keys (PKs), Foreign Keys (FKs), and integrity constraints.

3. To support all core library functionalities through efficient SQL queries and stored procedures.
4. To ensure data persistence and reliability for the LMS application.

5. Core Database Functionalities and Query Support

The DB must support the following essential management functions:

5.1 Book Management

- **Data Fields:** Book ID (PK), Title, Author, ISBN, Publisher, Total Copies, Available Copies.
- **Query Support:** Adding new books, updating copy counts, searching by Title, Author, or ISBN.

5.2 Member Management

- **Data Fields:** Member ID (PK), Name, Contact Info (Phone/Email), Member Type (Student/Faculty), Department.
- **Query Support:** Registering new members, updating contact details, retrieving borrowing history.

5.3 Issue, Return & Fine Management

- **Data Fields (Issue/Return Table):** Issue_ID (PK), Member_ID (FK), Book_ID (FK), Date_of_Issue, Due_Date, Return_Date (nullable).
- **Data Fields (Fine Management Table):** Fine_ID (PK), Issue_ID (FK), Days_Late, Fine_Amount_Calculated, Payment_Status.
- **FINES (Payment Records - Optional/Recommended):** Payment_ID (PK), Fine_ID (FK), Date_Paid, Amount_Paid. Separating Fine payment records from the calculation promotes better reporting and transactional integrity.
- **Query Support:**
  - Issuing a book (record transaction, decrement available copies).
  - Returning a book (update transaction, calculate fine, increment available copies).
  - Generating a report of all currently overdue books.
  - Calculating the total outstanding fine for a specific member.

6. Entity-Relationship Model (Conceptual Design)

The system will be based on the following primary entities and relationships: [Image of a Library Management System ERD]

Table 1: Conceptual Model Relationships (Crow's Foot)

Relationship	**Cardinality and Participation**
MEMBER to **ISSUE/RETURN**	One Member can participate in Many Issues/Returns ($1 : M$).
BOOK to **ISSUE/RETURN**	One Book (Master Record) can be involved in Many Issues/Returns ($1 : M$).
ISSUE/RETURN (Associative)	Links one Member to one Book at a specific time.

7. Tools & Technologies

- **RDBMS:** Oracle SQL (Specify one, e.g., Oracle SQL)
- **Modeling Tool:** Draw.io / MS Visio (for ERD generation)
- **Query Language:** Structured Query Language (SQL)
- **Integration:** The C++ application will interface with the DB using an appropriate connector .

8. Methodology and Deliverables

1. **Phase 1: Conceptual Design** (ERD, Normalization up to 3NF).
2. **Phase 2: Logical Design** (Schema definition, data types, PK/FK constraints).
3. **Phase 3: Physical Implementation** (Database creation in RDBMS).
4. **Phase 4: Functional Testing** (Demonstrating all required queries and integrity checks).

9. Conclusion

The proposed Library Management Database System provides the necessary structural integrity and efficient retrieval capabilities required to support a scalable and reliable digital library operation. This project will serve as a practical demonstration of core Database Management System principles, ensuring the LMS application has a robust, persistent back-end.